

TaxaWizard

Conversational workflow designer for the TaxaID ecosystem. An LLM-powered interview identifies your inputs, parameters, and pipeline steps, then generates a self-contained R script, methods text, or Shiny application. Although one could use other LLMs to do the same, TaxaWizard comes pre-informed with context about the TaxaID package.

The TaxaID ecosystem has many possible workflows, and choosing the right combination of packages and parameters can be daunting. TaxaWizard makes this easier through a guided conversation: describe your data and goals, and it generates the appropriate script. `workflow_app()` can also convert any R script (not just TaxaID scripts) into a Shiny application with point-and-click inputs.

Installation

```
install.packages(c("httr2", "jsonlite")) # required
install.packages("shiny")                # optional, for browser/viewer mode
devtools::install("path/to/TaxaWizard")
```

Requires an Application Programming Interface (API) key for an LLM provider (Anthropic Claude – Anthropic PBC, San Francisco, California – by default; Google Gemini, OpenAI, or local Ollama are also supported via TaxaTools). See the TaxaTools API Setup vignette for configuration.

Quick Start

```
library(TaxaWizard)

# Launch the interview (auto-selects console, browser, or RStudio viewer)
workflow_create()

# Or specify a mode
workflow_create(mode = "browser") # opens in web browser
workflow_create(mode = "console") # readline in terminal
workflow_create(mode = "viewer")  # RStudio Viewer pane
```

The assistant asks about your data type, available inputs, and analysis goals, then generates a script tailored to your path through the TaxaID pipeline. Supported workflow paths include:

- **eDNA / metabarcoding** — DADA2 seqtab or FASTA → BLAST (Basic Local Alignment Search Tool; Altschul et al. 1990, hosted by NCBI, the National Center for Biotechnology Information, U.S. National Library of Medicine, National Institutes of Health, Bethesda, Maryland) → likelihood model → Bayesian or LLM assignment
- **Acoustic** — BirdNET-Analyzer (Cornell Lab of Ornithology, Cornell University, Ithaca, New York) CSV output → match data; or Xeno-canto (Xeno-canto Foundation, Netherlands) reference recordings + BirdNET → acoustic likelihood model
- **Camera trap / image** — `animl` (Conservation Technology Lab, San Diego Zoo Wildlife Alliance, San Diego, California), `iNaturalist CV` (a joint initiative of the California Academy of Sciences and the National Geographic Society, San Francisco, California), or `SpeciesNet` (Google LLC, Mountain View, California) output → match data; or labeled reference images → image likelihood model
- **Reference library building** — taxa names (from TaxaExpect or user-supplied) → site-specific NCBI reference library; or load a local CRABS or FASTA database

- **Occurrence-based priors** — taxa + location → GBIF (Global Biodiversity Information Facility; GBIF Secretariat, Copenhagen, Denmark) occurrences → habitat → spatially explicit priors
- **Assignment convergence** — any combination of the above → likelihoods + priors → posteriors → consensus → report

When the interview is complete, TaxaWizard generates:

- **R script** with checkpoint/resume, error recovery, and debug mode
- **Methods text** summarizing the workflow for manuscripts
- **Shiny app** (via `workflow_app()`) for point-and-click execution

Fixing Errors in Generated Scripts

```
# If a generated script errors, resume the conversation with context
workflow_fix()
```

Converting Scripts to Shiny Apps

```
# Convert any TaxaID-generated (or annotated) script to a Shiny app that your clients could use to do r
workflow_app("my_workflow.R")

# Annotate a generic R script first, then convert
annotate_script("my_analysis.R")
workflow_app("my_analysis.R")
```

Key Functions

Function	Purpose
<code>workflow_create()</code>	Launch interactive interview (main entry point)
<code>workflow_fix()</code>	Resume after script error with diagnostic context
<code>workflow_app()</code>	Convert generated script to Shiny app
<code>annotate_script()</code>	Annotate generic R scripts for Shiny conversion
<code>workflow_engine()</code>	Stateless LLM engine (advanced / programmatic use)

Part of TaxaID

TaxaWizard sits outside the TaxaID dependency chain. It reads package metadata files and generates scripts that call the other TaxaID packages – it does not import them directly.

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, <https://doi.org/10.5066/xxxxxx>.

Software Requirements

- R ($\geq 4.1.0$; R Core Team 2025)
- An LLM API key (Anthropic recommended) is required for the conversational engine
- shiny (for browser/viewer chat interface and generated apps; in Suggests)

All dependencies are declared in the DESCRIPTION file and installed automatically.

Developed with Claude Code (Anthropic PBC, San Francisco, California).

References

Altschul, S.F., Gish, W., Miller, W., Myers, E.W. and Lipman, D.J. (1990). Basic local alignment search tool. *Journal of Molecular Biology*, 215(3), 403–410.

R Core Team (2025). R: A Language and Environment for Statistical Computing. V.4.5.2. R Foundation for Statistical Computing, Vienna, Austria. <https://www.r-project.org>